

AAYUSH ACHARYA

 aayushacharya.com.np  acharya.aayush.647@gmail.com  linkedin.com/in/aayush-acharya  [aayushacharya](https://github.com/aayushacharya)

Education

Temple University

2024 – Present

PhD in Computer Science(transferred from MS in Spring 2025)

MS in Computer Science(Fall 2024)

Tribhuvan University, IOE Pulchowk Campus

2017 – 2022

Bachelor in Computer Engineering

Research

Temple University

2024 – Present

Graduate Research Assistant

- Conducting research on LLM-based information extraction for **automated Climate Knowledge Graph Construction** with a focus on **climate models from ESGF**. [Website](#)  — [Knowledge Graph](#) 
- Developing and evaluating **LLM-driven subgraph retrieval** methods to improve graph-aware reasoning and retrieval accuracy. [GitHub Repository](#) 
- Investigating **natural-language-to-graph-query translation** approaches (Cypher generation) to enable intuitive access to structured graph data. [ChatBot](#) 

Nepal Applied Mathematics and Informatics Institute for research (NAAMII)

2021 – 2022

Research intern

- Implemented a seamless integration of **Automatic Speech Recognition (ASR)** and Natural Language Processing (NLP) by employing the self-supervised wav2vec2 model for audio transcription. The integration enhances the overall system's accuracy and efficiency in processing **Nepali audio data**. [Thesis](#) 
- Integrated a BERT-based Multilingual Representations from Indian Languages (MuRIL) model specifically designed for Indic languages like Nepali. This incorporation significantly enhances the system's ability to understand and process the nuances of the Nepali language, ensuring more accurate **categorization of the audio content**.

Professional experience

Rara Labs

2022 – 2024

Software Engineer

- Led diverse projects in healthcare and fintech sectors, specializing in **software engineering** using Golang and **machine learning** with Python.
- Wrote **CI/CD** pipelines and integration testing frameworks, ensuring seamless software delivery and quality assurance.
- Started and **led biweekly knowledge-sharing sessions** in the company, fostering team collaboration and exchange of new findings and collaboratively solving challenges.
- Provided mentorship to undergraduate students and successfully **promoted to a Project Lead**, recognized for leadership and technical proficiency.

Open source contributions

georgia-tech-db/evadb

2.5k GitHub stars

Database system for AI-powered apps

- Integrated arXiv repository data as part of third party databases in EvaDB. This allows performing SQL queries on the metadata and content of papers present in arXiv. [Pull Request](#) 
- Integrated Google's Gemini generative model for chat completion. The example usage of this functionality may be to generate labels from a video summarizing model's output in EvaDB. [Pull Request](#) 
- Added support for arithmetic expressions in projection with precedence in a lark based query parser. This allowed execution of queries like `SELECT 1+2`; [Pull Request](#) 

go-gorm/gorm

34.4k GitHub stars

Popular ORM in Golang.

- Fixed an issue which allowed new database session to be created with trigger hooks disabled. [Pull Request](#) 

Teaching/Mentoring experience

Temple University

2024 – Present

Graduate Teaching Assistant

- Teaching assistant for Discrete Mathematics for three semesters
- Taught two sections in each semester, held recitation classes, held weekly office hours and graded quizzes.

IOE Pulchowk Campus, Tribhuvan University in collaboration with Rara Labs

2022 – 2023

Undergraduate thesis mentor

Presentations

7th International Workshop on Effective Engineering Education (IWEEE) — *Junior year*

2022

- Presented an **image classification model** designed to assist farmers by enabling them to **capture pictures of pests**, identify them, and suggest their natural control methods.

Notable projects

HMS with Early Warning Sepsis Prediction System | *Python, Go*

- Utilized patient data, including demographics and EMR records, to build **predictive models** for **early sepsis detection**.
- Developed a **healthcare management** system featuring patient allocation across hospitals, wards, and beds.

Cross border Request to Pay(RTP) System | *Go*

- **Created CI/CD pipelines**, with automated **integration testing** and **deployment**, reducing failures in production system.
- Implemented an **event-driven architecture** leveraging RabbitMQ to facilitate real-time data processing between microservices.

Affiliations

Rara Labs: Software Engineer — *2022-2024*

Nepal Applied Mathematics and Informatics Institute for research (NAAMII): Research intern — *2021-2022*

Logpoint Inc.: Solutions intern — *2022-2022*

Microsoft Student Learn Ambassador(MLSA): Beta Member — *2020-2022*

Engineering Youth Network Amnesty International Nepal: Member — *2019-2022*

Zerone Magazine: Editor — *2021-2022*

Technical skills

Python, PyTorch, Golang, Neo4J, AWS, Azure, GitHub Actions, Docker, Flask, Node.js, RabbitMQ, React, Typescript, Git, Linux, Shell

Honors and outreach

- Ranked under top 1% in undergraduate entrance examination among 15000 students.
- Regular receiver of TU stipend, a merit based semester wise allowance for academically well-performing students.
- Ranked 3rd out of 600 students in high school in National level examinations.
- Organized LOCUS Hack-a-week, a 7-day hackathon event with nationwide reach and > 50 applying teams.
- Organized Advanced C workshop along with a workshop on Particle Swarm Optimization as part of MLSA.
- Blog with over 1.6K unique visitors per month according to Cloudflare analytics(as of Jul 2024).

References

Longin Jan Latecki: Professor at Temple University. Email: *latecki@temple.edu*